

Credit risk prediction project

- **Project Idea**

Building a *Credit Risk* model to predict the probability of customer default within two years using financial and behavioral data, in order to support lending decisions and reduce financial losses.

- **Project Plan**

P1: Vision + Data Collection

- Define the problem: Credit Default Prediction
- Define the objective: reduce financial losses and improve lending decisions
- Understand the target variable (SeriousDlqin2yrs)
- Load and verify data quality

P2: EDA + Feature Engineering

- Analyze data distributions
- Understand relationships between features and target
- Identify patterns (e.g., late payments increase default risk)
- Create new features such as:
 - Total number of late payments
 - Debt-to-income variations

P3: Preprocessing + Imbalance Handling

- Handle missing values (e.g., MonthlyIncome, NumberOfDependents)
- Treat outliers in financial and delay features
- Apply scaling/normalization if needed
- Handle class imbalance using:
 - SMOTE or class weights

P4: Modeling + Tuning

- Build a baseline model (Logistic Regression)
- Train advanced models:
 - Random Forest
 - XGBoost / LightGBM
- Perform hyperparameter tuning (Grid Search / Random Search)

P5: Evaluation + Deployment

- Evaluate using:
 - Recall (primary metric)
 - ROC-AUC
- Confusion matrix analysis + threshold tuning
- Convert probabilities into decisions (approve/reject)
- Optional deployment:
 - API (Flask/FastAPI) or dashboard

- **Roles**

1) Malak Adel (Data Engineer (Data Preparation & Preprocessing)):

Responsibilities:

Acquire and load the dataset

Perform data validation and consistency checks

Handle missing values using appropriate strategies

Detect and treat outliers in financial variables

Apply preprocessing techniques (scaling, normalization, encoding)

Prepare a clean and reliable dataset for downstream tasks

2)Arwa Mahmoud (Data Analyst (Exploratory Data Analysis - EDA)):

Responsibilities:

Conduct in-depth Exploratory Data Analysis (EDA)

Analyze feature distributions and detect anomalies

Study relationships between features and the target variable

Build visualizations (histograms, boxplots, correlation matrices)

Generate actionable insights to guide modeling decisions

3)Nour Hazem (Feature Engineer (Feature Engineering & Selection)):

Responsibilities:

Design and create new predictive features

Apply feature transformation techniques

Perform feature selection to reduce noise and redundancy

Implement dimensionality reduction methods (if needed)

Enhance data representation to improve model performance

4)Ahmed Abdelrazek [team leader] (Machine Learning Engineer (Baseline Modeling)

Responsibilities:

- Build advanced models (Random Forest, XGBoost, LightGBM)
- Perform hyperparameter tuning (Grid Search / Random Search)
- Handle class imbalance (e.g., SMOTE, class weighting)
- Optimize model performance
- Compare models and select the best candidate

5)Sama Mohamed Machine Learning Engineer (Advanced Modeling & Optimization)

Responsibilities:

- Build advanced models (Random Forest, XGBoost, LightGBM)
- Perform hyperparameter tuning (Grid Search / Random Search)
- Handle class imbalance (e.g., SMOTE, class weighting)
- Optimize model performance
- Compare models and select the best candidate

6)Mohamed Reda (Model Evaluation, Deployment & Reporting Specialist)

Responsibilities:

- Evaluate model performance using:
 - Recall (primary metric for risk detection)
 - ROC-AUC
 - Confusion Matrix
- Perform threshold tuning based on business needs
- Translate model outputs into decision rules (approve/reject)
- (Optional) Deploy the model via API or dashboard
- Prepare final documentation and project report

Literature Review (Similar Projects)

1) Kaggle – Give Me Some Credit

- A well-known dataset used for credit risk prediction.
- Objective: predict whether a customer will experience financial distress within 2 years.
- Common models:
 - Logistic Regression
 - Random Forest
 - XGBoost (most effective in many solutions)
- Key finding: payment delay and credit utilization are strong predictors of default risk.

2) Home Credit Default Risk (Kaggle Competition)

- A large-scale real-world credit scoring problem.
- Objective: predict whether a client will repay a loan.
- Approach:
 - Extensive feature engineering across multiple tables
 - Gradient boosting models (LightGBM, XGBoost)
- Key finding: engineered features significantly improve model performance.

3) Lending Club Loan Default Prediction

- Dataset from a peer-to-peer lending platform.
- Objective: predict loan default before approval.
- Models used:
 - Logistic Regression
 - Decision Trees
 - Ensemble methods
- Key finding: income level, interest rate, and credit history are strong risk indicators.

4) German Credit Dataset (Academic Study)

- Classic dataset widely used in academic research for credit scoring.
- Objective: classify customers into good or bad credit risk.
- Models used:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Trees
 - Neural Networks
- Key finding: simpler models like Logistic Regression remain competitive due to interpretability in financial applications

- **Requirements (System Definition)**

Users

- *Credit Officer / Loan Approver***

- Uses the system to evaluate customer risk before approving loans.

- *Risk Analyst / Data Scientist***

- Monitors model performance and improves features/models.

- *System (ML Model)***

- Automatically processes input data and produces risk predictions.

System Functional Requirements

1) Input Layer:

The system should accept customer financial and behavioral data, such as:

- Age
- Monthly income
- Debt ratio
- Credit utilization ratio
- Number of credit lines and loans
- Payment delay history (30–59, 60–89, 90+ days)
- Number of dependents

2) Data Processing

The system must:

- Handle missing values (e.g., income, dependents)
- Detect and treat outliers
- Transform and encode features if needed
- Scale numerical variables when required

3) Risk Prediction

The system should:

- Output a *probability of default (0 to 1)* for each customer
- Classify customers into:
 - Low Risk
 - Medium Risk
 - High Risk

4) Decision Support

Based on the predicted probability, the system should:

- Approve loan
 - Reject loan
 - Flag for manual review
- (Decision is based on a predefined threshold)

5) Model Evaluation

The system performance should be evaluated using:

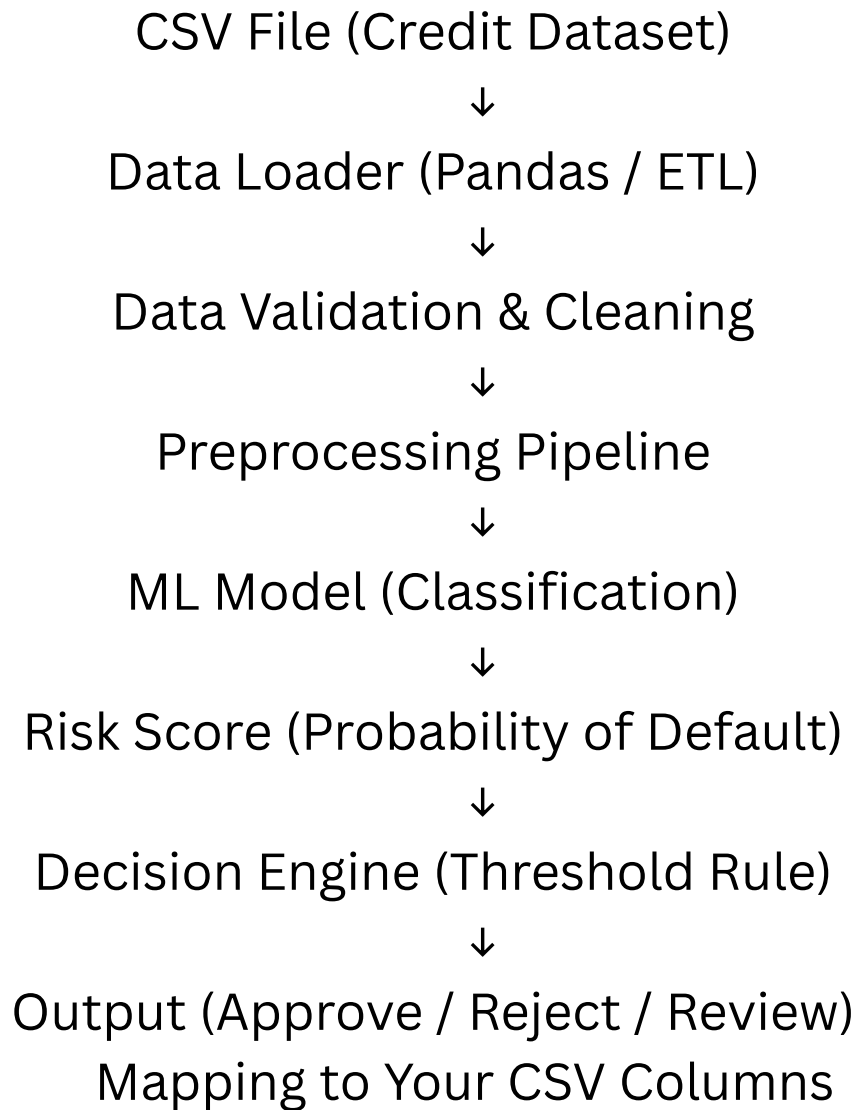
- *Recall* (to capture high-risk customers)
- *ROC-AUC* (overall classification quality)
- Confusion Matrix analysis

6) Explainability (Optional but recommended)

The system should provide:

- Feature importance
 - Model explanations (e.g., SHAP values)
- to support transparency in financial decisions

- **System Architecture**



Input Features:

RevolvingUtilizationOfUnsecuredLines
age
NumberOfTime30-59DaysPastDueNotWorse
NumberOfTime60-89DaysPastDueNotWorse
NumberOfTimes90DaysLate
DebtRatio
MonthlyIncome
NumberOfOpenCreditLinesAndLoans
NumberRealEstateLoansOrLines
NumberOfDependent

Target:

SeriousDlqin2yrs (1 = default, 0 = no default)

- **UML Design**

Use Case Diagram

Credit Analyst

|

|--> Upload CSV file

|--> View predictions

|--> Analyze risk results

System

|--> Load CSV data

|--> Clean and preprocess data

|--> Run ML model

|--> Return risk score

Class Diagram

CSVLoader

- load_data()

Preprocessor

- handle_missing_values()

- handle_outliers()

- scale_features()

CreditRiskModel

- train()

- predict_proba()

DecisionEngine

- threshold = 0.7
- classify_risk()

Sequence Diagram

User → System: Upload CSV file

System → Loader: Read dataset

Loader → Preprocessor: Clean data

Preprocessor → Model: Processed features

Model → System: Return probabilities

System → Decision Engine: Apply threshold

Decision Engine → User: Return final risk results

Dataset Link

Data Format

File type: CSV (Comma-Separated Values)

Structure: Tabular dataset (each row = one customer)

Task: Binary classification problem

Columns

Target

SeriousDlqin2yrs (0 = no default, 1 = default within 2 years)

Features

RevolvingUtilizationOfUnsecuredLines

age

NumberOfTime30-59DaysPastDueNotWorse

NumberOfTime60-89DaysPastDueNotWorse

NumberOfTimes90DaysLate

DebtRatio

MonthlyIncome

NumberOfOpenCreditLinesAndLoans

NumberRealEstateLoansOrLines

NumberOfDependents

Summary

Dataset type: CSV tabular data

Each row: one loan applicant

Features: 10 input variables